Three Copies, Which Is Right?

Replication and consistency

SCAN / LEARN

THE MAIN IDEA

Replication stores copies of data in different places. Copies may temporarily disagree, so the system needs rules for updates, reads and conflicts.

FOLLOW THE SYSTEM

01 ORIGINAL

A customer changes an address.

02 PROPAGATE

The update travels to other replicas.

03 DELAY

One replica still has the old address.

04 RECONCILE

A consistency rule determines what to return.

WRITE → COPY → DELAY → CONSISTENCY RULE

COMMON MISCONCEPTION

Three copies always contain identical data at every instant.

MORE ACCURATE

Some designs allow temporary differences to remain available.

THREE QUESTIONS TO KEEP

PROBLEM: How do copies stay useful after an update?

FAILURE: Stale information produces wrong decisions.

IN REAL LIFE: Cloud databases and global services.

SOLO DISCOVERY

Write two different updates on two sheets. Which wins, and why?